Take a positive integer. If it is even, halve it; if it is odd, triple it and add one. Repeat.

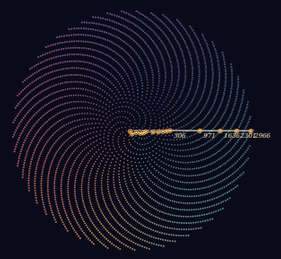
The Collatz conjecture says that every starting value eventually reaches 1. It can be explained to a child in a minute and has resisted every attempt at proof since the 1930s. This book is about why.

Reading the map as a random walk, an optimizer and a Diophantine problem, the authors split the conjecture into a divergence half, a law of large numbers for a single deterministic orbit, and a cycle half, a question about rational approximation of $\log_2 3$ that Diophantine information alone cannot finish. Along the way they prove that no Lyapunov function increasing in the number certifies descent over any fixed number of steps and that no finite residue sieve certifies descent for every integer, and they set out a computational path to the problem with its limits made precise.

The arithmetic of the map also yields an algorithm. Switching MM, a first order method with momentum $\log_2 3 - 1$ applied on alternate steps, has an explicit linear rate at step $1/L$ on smooth strongly convex functions, and on quadratics with additive gradient noise, at steps up to $1/L$, its noise variance stays below $5/3$ times that of gradient descent for every momentum coefficient below one.

With full proofs, open problems and code for every number, this is a book for probabilists, number theorists and optimizers alike.

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On the Collatz Conjecture

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Martingales, Momentum and Continued Fractions

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